# Mapping

**BU-ISCIII**

**Unidades Comunes Científico Técnicas – SGSAFI-ISCIII**

26-30 Octubre 2026, 6ª Edición
Programa Formación Continua, ISCIII

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**Mapping against reference genome and Variant Calling :**

- Mapping vs Alignment
- What is mapping?
- How to choose a NGS mapper.
- SAM/BAM format
- Duplicate filter

# Alignment

## Definition:

Arrange two or more nucleotide or aminoacid sequences to identify regions of similarity that may be a consequence of functional, structural, or evolutionary relationships.

```
AAB24882        TYHMCQFHCRYVNNHSGEKLYECNERSKAFSCPSHLQCHKRRQIGEKTHEHNQCGKAFPT
AAB24881        --------------------YECNQCGKAFAQHSSLKCHYRTHIGEKPYECNQCGKAFSK

AAB24882        PSHLQYHERTHTGEKPYECHQCGQAFKKCSLLQRHKRTHTGEKPYE-CNQCGKAFAQ-
AAB24881        HSHLQCHKRTHTGEKPYECNQCGKAFSQHGLLQRHKRTHTGEKPYMNVINMVKPLHNS
```

# Multiple alignment (MSA)

## Definition:

A multiple alignment is a collection of three or more sequences partial or completely aligned.



# Mapping definition

## Definición:

Place a sequence inside a larger sequence. For example, determine the position of a read inside a reference genome.



# Alignment vs mapping

## Mapping:

- A mapping is regarded to be correct if it overlaps the true region.
- Each read maps independently
- From thousand to millions of sequences.

## Multiple alignment:

- An alignment is regarded to be correct only if each base is placed correctly.
- Minimizes differences among sequences
- From tens to hundred of sequences.

## Consideratiosn:

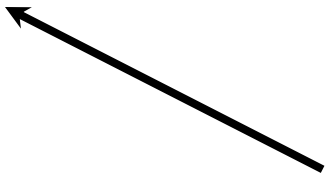
An algorithm can be good at mapping but may not be good aligning.

This is because the true alignment minimizes differences between reads, but the read mapper only sees the reference.

Hen Li. Mapping, Alignment and SNP Calling. MPG Next Gen Workshop 2011

# So in summary…

CTGACCTCATGTGATCCACCCGCCTTGGC
C



Find best match for the read
in a reference sequence

TGATCCAC

## Challenges

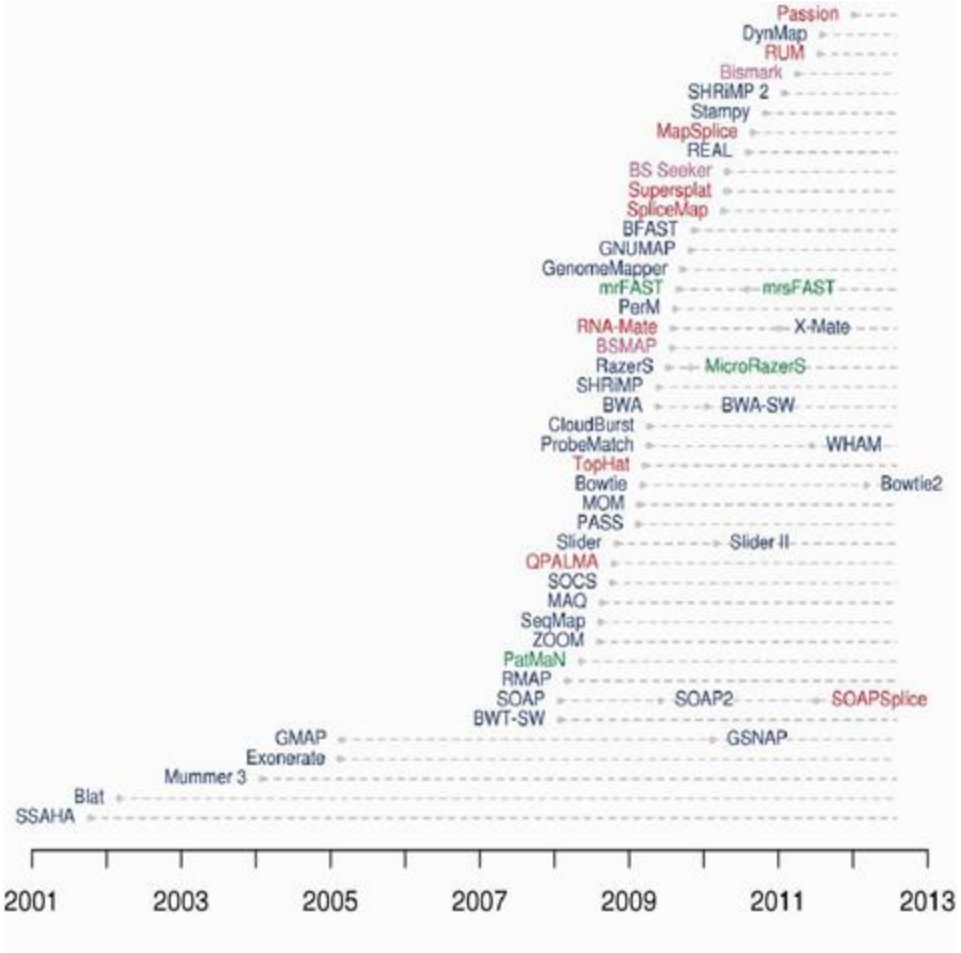
- Errors in reads
- Errors in libraries
- Repetitive regions (repeats, homologous regions)
- Homopolymers
- Individual polymorphisms

Pierre Lechat. Variants Calling lecture. Pasteur.fr

# What mapper should I use?

## Mappers:

- Más de 60 mappers available.
- Lots of papers reviewing its performamnce.



# What mapper should I use?

## Things to consider:

- Computational resources vs sensibility
- Platform and type of experiment (Illumina/454/etc,paired-end,DNA/RNA/etc)
- Variation (indels allowance, mistmatch number,etc.)
- Repetitions (all regions, best match, random, user

## Important:

Default options don't have to be the best:

"… there is no tool that outperforms all of the others in all the tests. Therefore, the end user should clearly specify his needs in order to choose the tool that provides the best results." - Hatem et al *BMC Bioinformatics* 2013, **14**:184

# End-to-end vs local alignment

## End-to-end

```
Read:      GACTGGGCGATCTCGACTTCG
Reference: GACTGCGATCTCGACATCG

Alignment:
  Read:      GACTGGGCGATCTCGACTTCG
             |||||  |||||||||| |||
  Reference: GACTG--CGATCTCGACATCG
```

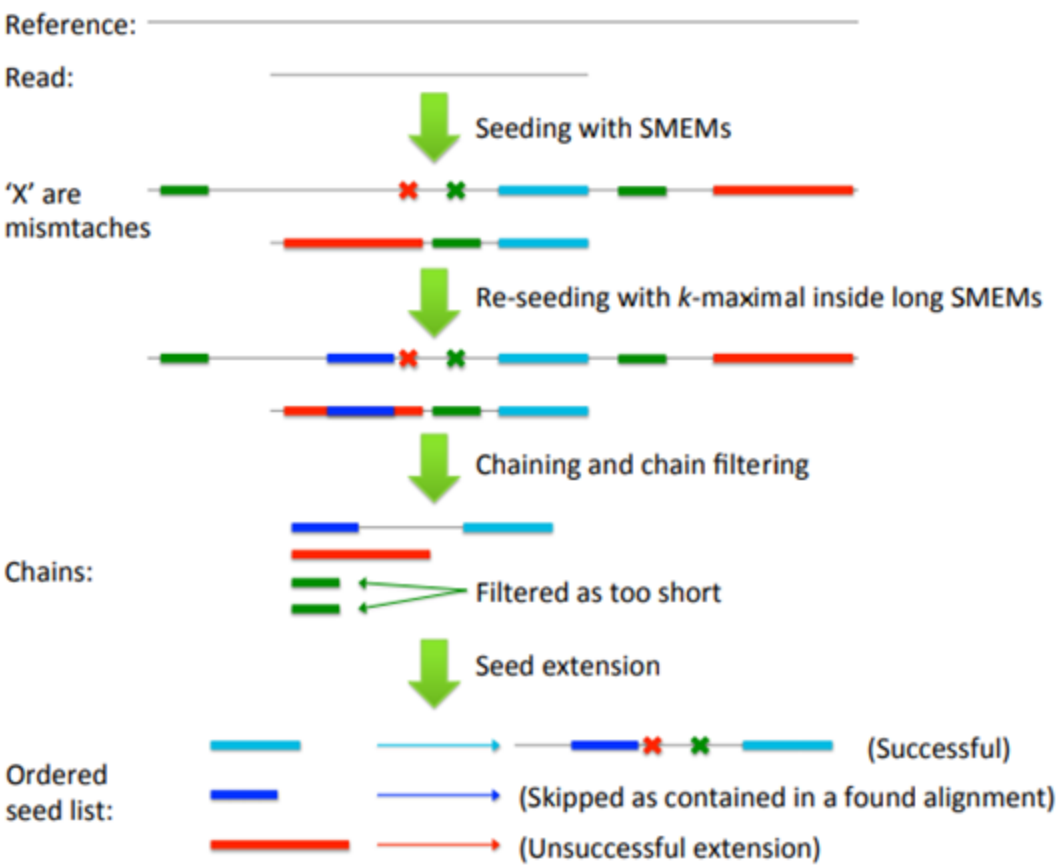
## Local

```
Read:      ACGGTTGCGTTAATCCGCCACG
Reference: TAACTTGCGTTAAATCCGCCTGG

Alignment:
  Read:      ACGGTTGCGTTAA-TCCGCCACG
                 ||||||||| ||||||
  Reference: TAACTTGCGTTAAATCCGCCTGG
```

Bowtie2 manual.

# BWA MEM




## SMEM strategy

- Maximal exact match (MEM): an exact match that cannot be extended further in either direction
- Super-maximal exact match (SMEM): a MEM that is not contained in any other MEMs on the query coordinate (Li, 2012). At any query position, the longest exact match covering the position must be a SMEM.

## Seed-and-extend algorithm

## Local alignment

Hen LI. Aligning sequence reads, clone sequences and assembly con*gs with BWA-MEM. Poster. Broad Institute.

# BOWTIE2

**End-to-end alignment by default.**

**Three reporting modes:**

- Best alignment
- K alignments
- All alignments

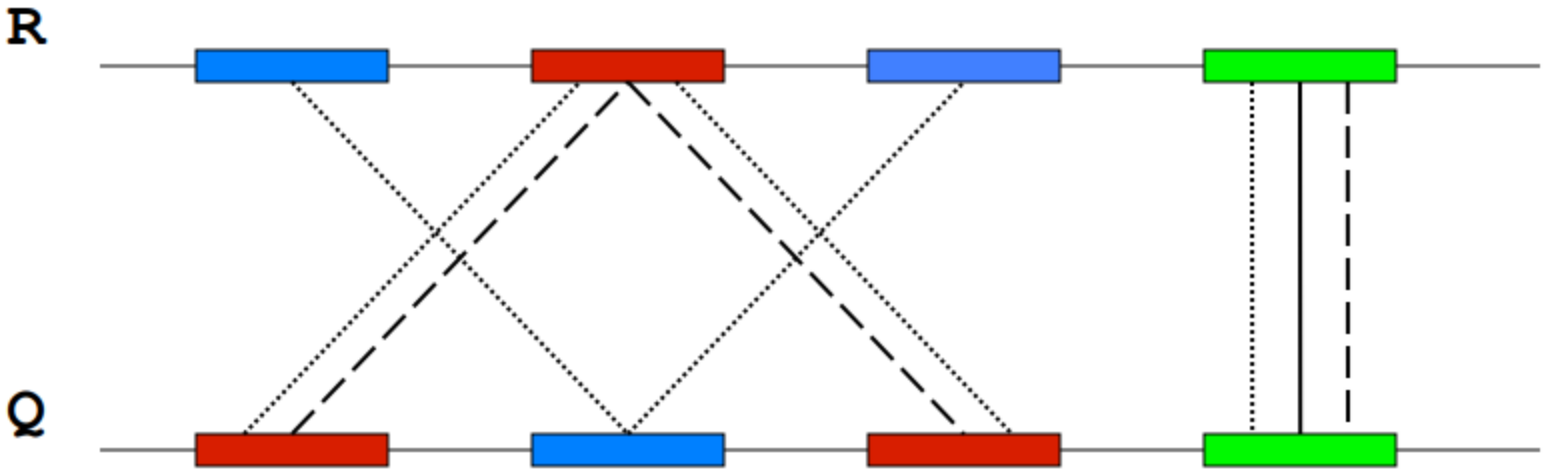
**Lots of customizable parameters that change its performance.**

# Example whole genome aligner: MUMMER

**MUM** : maximal unique match

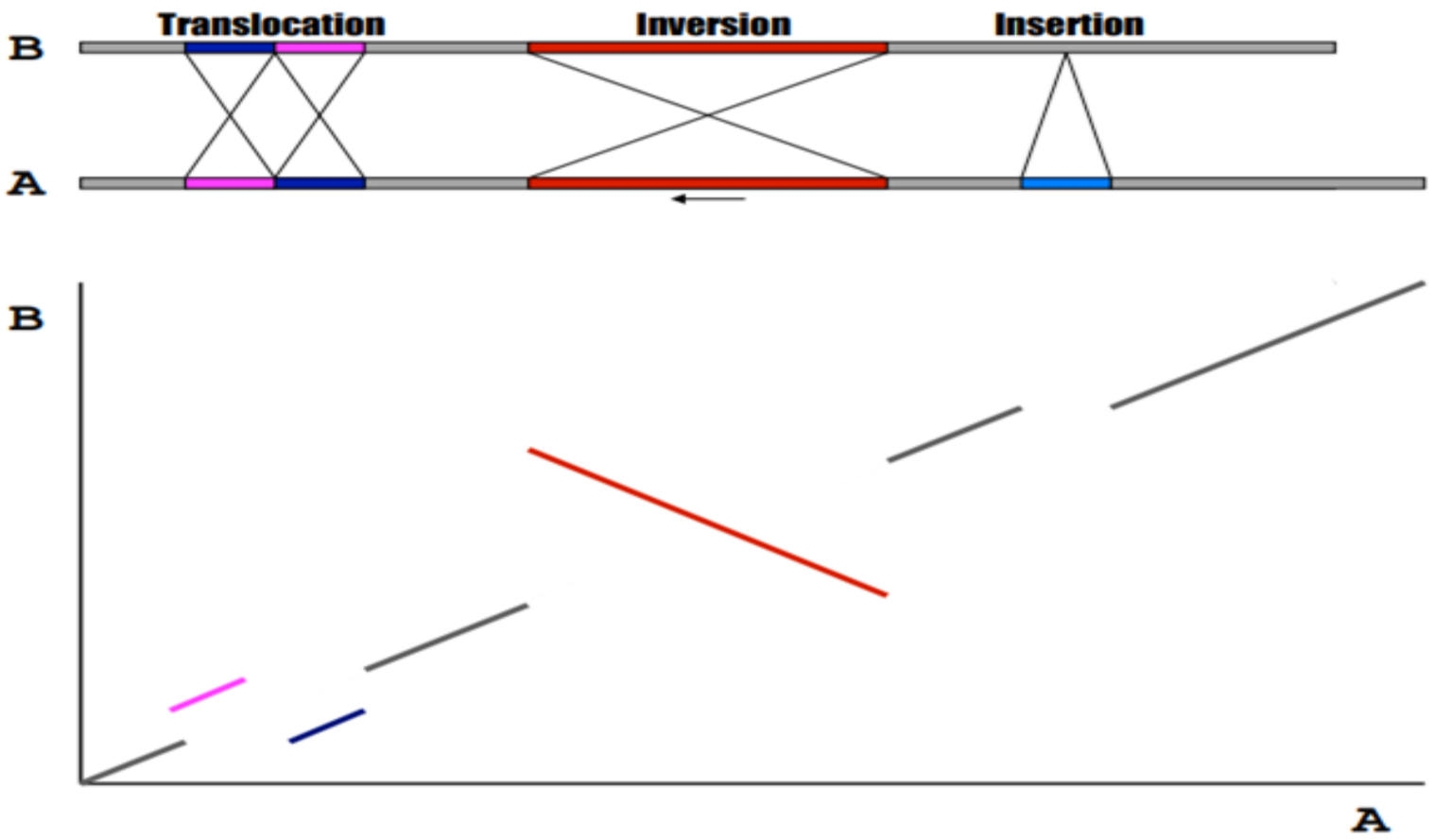
**MAM** : maximal almost-unique match

**MEM** : maximal exact match



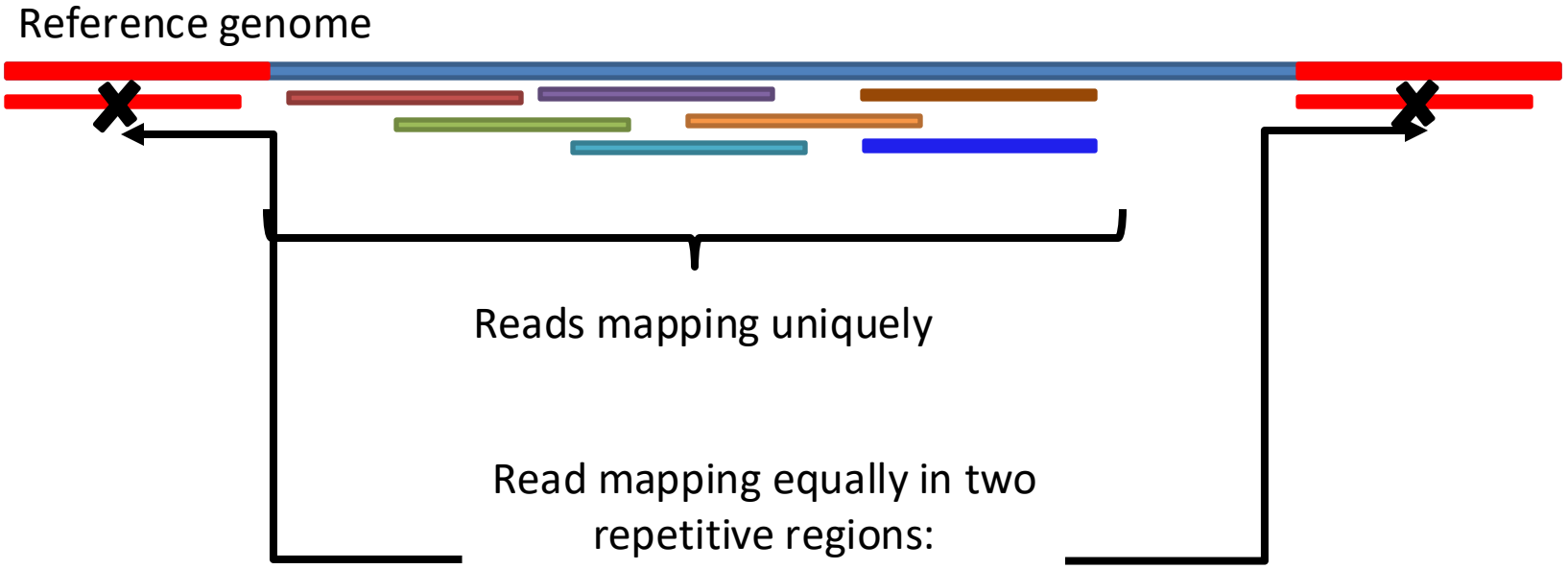

Adam M. Phillippy. Whole Genome Alignment with MUMmer. Presentation.

# Example whole genome aligner: MUMMER




Adam M. Phillippy. Whole Genome Alignment with MUMmer. Lecture.

# Which aligner should I use for aligning reads agains a complete genome for variant calling?

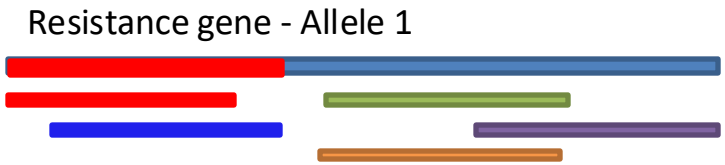


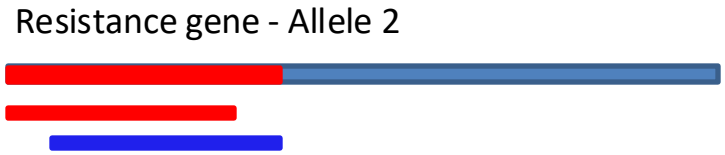
# Which aligner should I use for aligning reads against a resistance gene database for determining with resistance genes I have in my sample?



Homologus/repetitive region

Reads mapping to the repetitive/homologus region map against all alleles.
**We** allow one read to map to **several locations**.





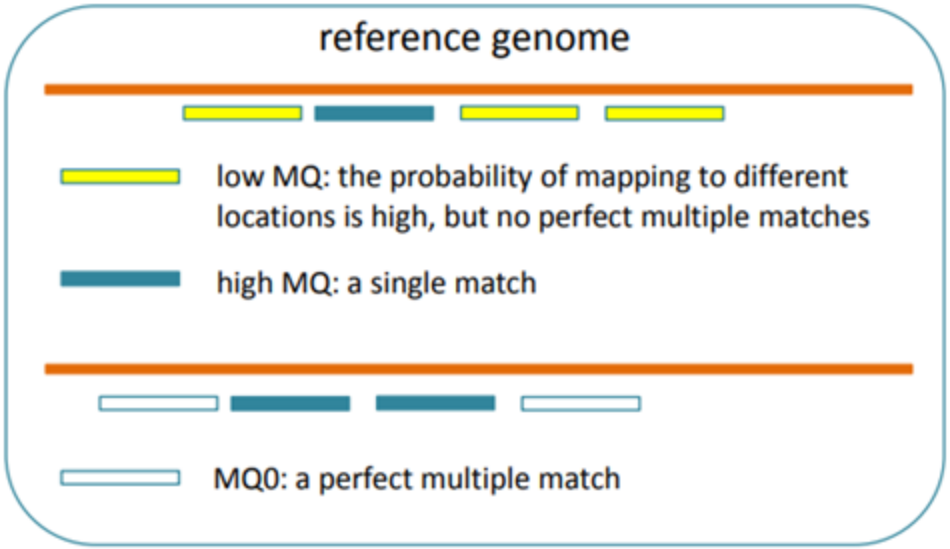


Reads mapping uniquely only map in Allele 1. Which is the one more **covered**

# MAPQ

- What if there are several possible places to align your sequencing read? This may be due to:
  - Repeated elements in the genome
  - Low complexity sequences
  - Reference errors and gaps

  **MQ is a phredScore of the quality of the alignment**



# MAPQ

**MAPQ is NOT comparable among mappers.**

**BWA:**

- MAPQ represents the probability of the read to be mapped correctly.
- MAPQ = 0 identifies unmapped reads and…

**Reads mapping to different locations!**

**BOWTIE2:**

- MAPQ represents the "uniqueness" of the read. A MAPQ < 10 indicates that there is at least a 1 in 10 chance that the read truly originated elsewhere
- MAPQ = 0 identifies unmapped reads

# SAM format

## Definición:

It's a specification that defines a generic format for storing nucleotide alignments. It describes a query alignment against a reference genome.

```
@HD VN:1.5 SO:coordinate
@SQ SN:ref LN:45
r001    99 ref  7 30 8M2I4M1D3M = 37  39 TTAGATAAAGGATACTG *
r002     0 ref  9 30 3S6M1P1I4M *  0   0 AAAAGATAAGGATA    *
r003     0 ref  9 30 5S6M       *  0   0 GCCTAAGCTAA       * SA:Z:ref,29,-,6H5M,17,0;
r004     0 ref 16 30 6M14N5M    *  0   0 ATAGCTTCAGC       *
r003  2064 ref 29 17 6H5M       *  0   0 TAGGC             * SA:Z:ref,9,+,5S6M,30,1;
r001   147 ref 37 30 9M         =  7 -39 CAGCGGCAT         * NM:i:1
```

# SAM format

| Col | Field | Type | Regexp/Range | Brief description |
|---|---|---|---|---|
| 1 | QNAME | String | `[!-?A-~]{1,255}` | Query template NAME |
| 2 | FLAG | Int | $[0,2^{16}-1]$ | bitwise FLAG |
| 3 | RNAME | String | `\*|[!-()+-<>-~][!-~]*` | Reference sequence NAME |
| 4 | POS | Int | $[0,2^{31}-1]$ | 1-based leftmost mapping POSition |
| 5 | MAPQ | Int | $[0,2^{8}-1]$ | MAPping Quality |
| 6 | CIGAR | String | `\*|([0-9]+[MIDNSHPX=])+` | CIGAR string |
| 7 | RNEXT | String | `\*|=|[!-()+-<>-~][!-~]*` | Ref. name of the mate/next read |
| 8 | PNEXT | Int | $[0,2^{31}-1]$ | Position of the mate/next read |
| 9 | TLEN | Int | $[-2^{31}+1,2^{31}-1]$ | observed Template LENgth |
| 10 | SEQ | String | `\*|[A-Za-z=.]+` | segment SEQuence |
| 11 | QUAL | String | `[!-~]+` | ASCII of Phred-scaled base QUALity+33 |

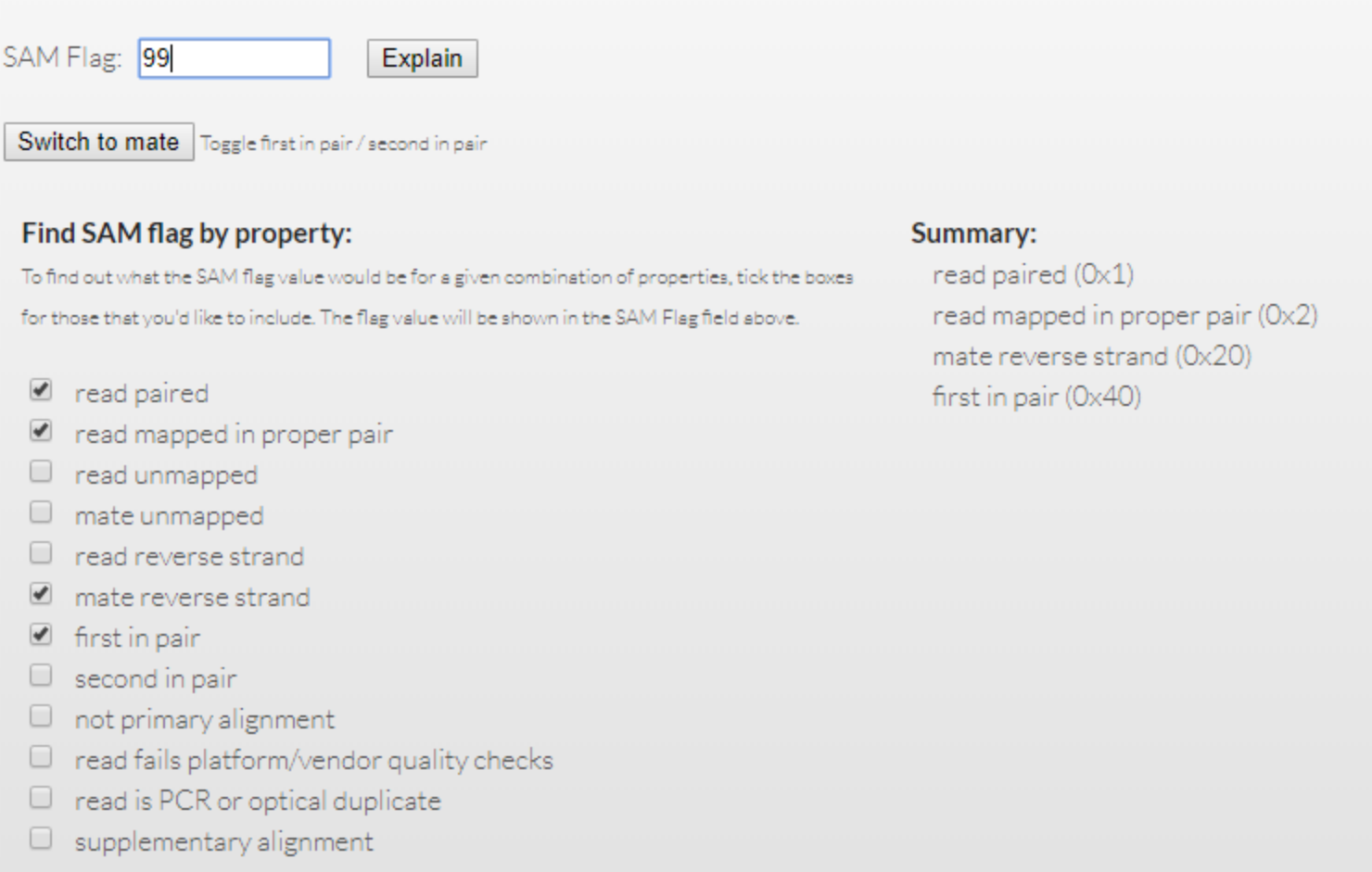
```
@HD VN:1.5 SO:coordinate
@SQ SN:ref LN:45
r001   99 ref  7 30 8M2I4M1D3M = 37  39 TTAGATAAAGGATACTG *
r002    0 ref  9 30 3S6M1P1I4M *  0   0 AAAAGATAAGGATA    *
r003    0 ref  9 30 5S6M       *  0   0 GCCTAAGCTAA       * SA:Z:ref,29,-,6H5M,17,0;
r004    0 ref 16 30 6M14N5M    *  0   0 ATAGCTTCAGC       *
r003 2064 ref 29 17 6H5M       *  0   0 TAGGC             * SA:Z:ref,9,+,5S6M,30,1;
r001  147 ref 37 30 9M         =  7 -39 CAGCGGCAT         * NM:i:1
```

# SAM format: flags

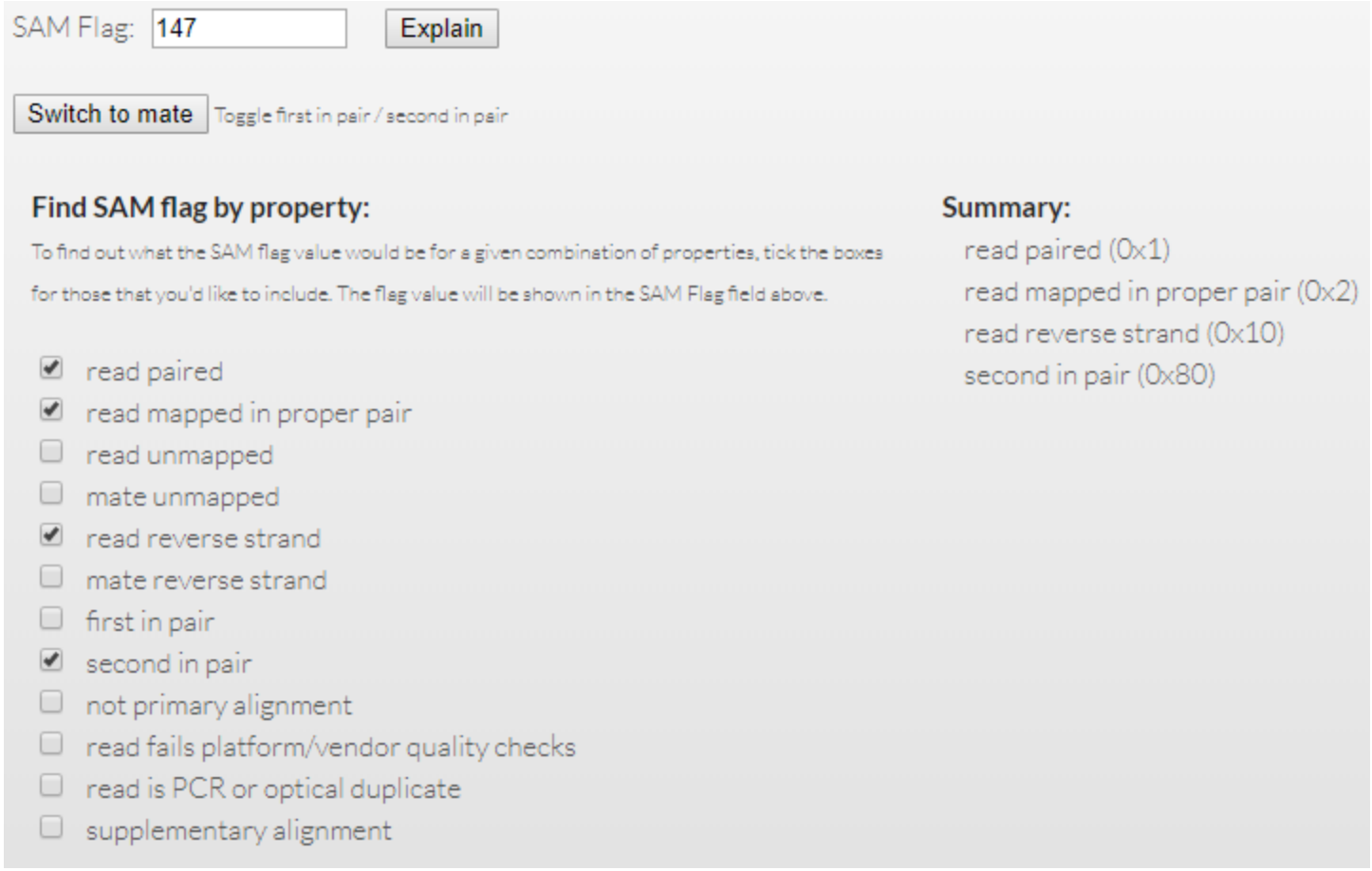
| Bit | Description |
|---|---|
| 0x1 | template having multiple segments in sequencing |
| 0x2 | each segment properly aligned according to the aligner |
| 0x4 | segment unmapped |
| 0x8 | next segment in the template unmapped |
| 0x10 | SEQ being reverse complemented |
| 0x20 | SEQ of the next segment in the template being reversed |
| 0x40 | the first segment in the template |
| 0x80 | the last segment in the template |
| 0x100 | secondary alignment |
| 0x200 | not passing quality controls |
| 0x400 | PCR or optical duplicate |
| 0x800 | supplementary alignment |

https://broadinstitute.github.io/picard/explain-flags.html

# Flag explanation example 1



# Flag explanation example 2



# SAM format: CIGAR string

| Op | BAM | Description |
|---|---|---|
| M | 0 | alignment match (can be a sequence match or mismatch) |
| I | 1 | insertion to the reference |
| D | 2 | deletion from the reference |
| N | 3 | skipped region from the reference |
| S | 4 | soft clipping (clipped sequences present in SEQ) |
| H | 5 | hard clipping (clipped sequences NOT present in SEQ) |
| P | 6 | padding (silent deletion from padded reference) |
| = | 7 | sequence match |
| X | 8 | sequence mismatch |

# SAM vs BAM format

- SAM and BAM format are exactly the same.
  - SAM is a tabular plain text file.
  - BAM is its binary format. Binary meaning is in a compress format not human readable.
  - We **MUST** always use BAM format because it is optimized for computer-reading

**AND**

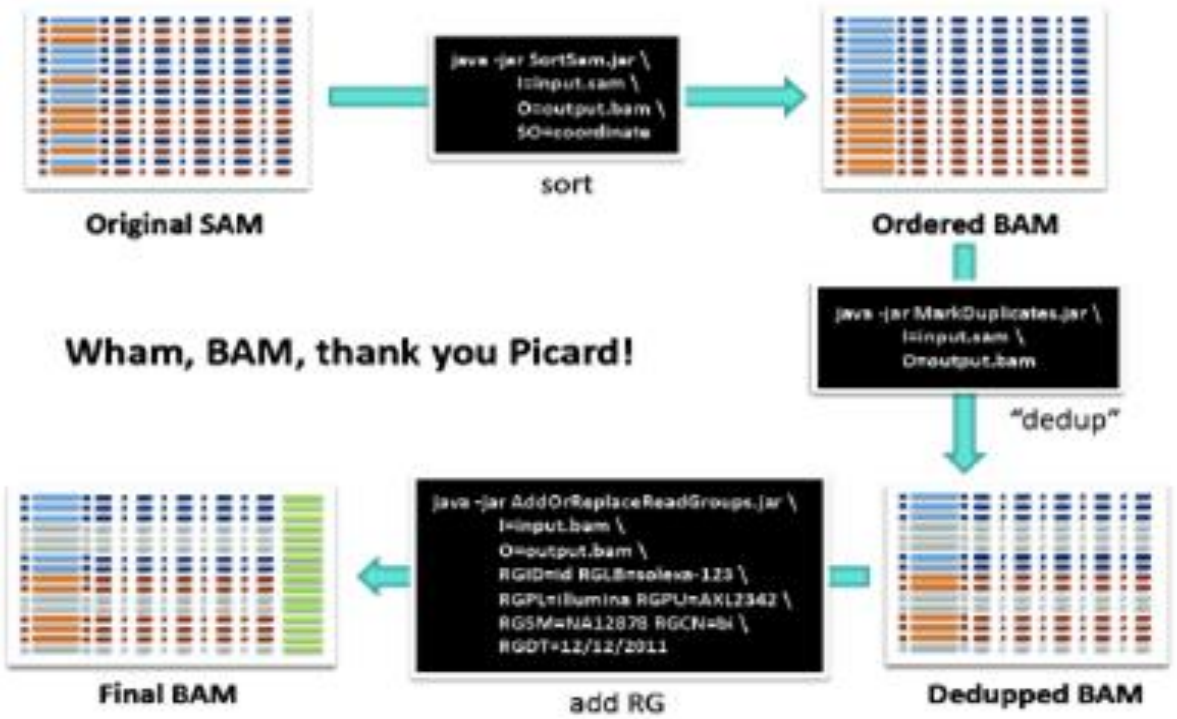
**BECAUSE IT SAVES A LOT OF DISK SPACE!!**

Typical bam and sam format files weights from a S. grumpensis
SAM format file: 3.6 GB
BAM format file: 689 M

# Duplicate filter

- Duplicates are non-independent measurements of a sequence
  - Sampled from the exact same template of DNA
  - Violates assumptions of variant calling
- Errors in sample/library prep will get propagated to all the duplicates
- Just pick the “best” copy – mitigates the effects of errors
- **Definition**: sequences starting and finishing in the exact same coordinates. Both pairs if paired-end.

# Duplicate filter



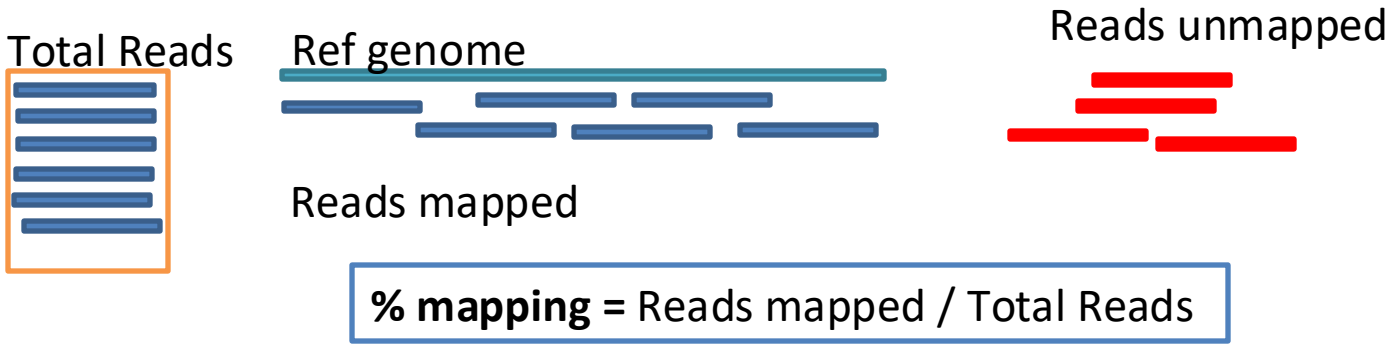
# Mapping statistics

- % mapped: reads mapped/total reads
- % unmapped: reads unmapped/total reads
- % duplicates: reads belonging to same template/total reads
- Mean depth of coverage
- Coverage: % genome with at least one read mapped.

# Mapping quality control

- **% mapping:** number of reads mapping againts reference genome.

Picard
Samtools




**% mapping** = Reads mapped / Total Reads

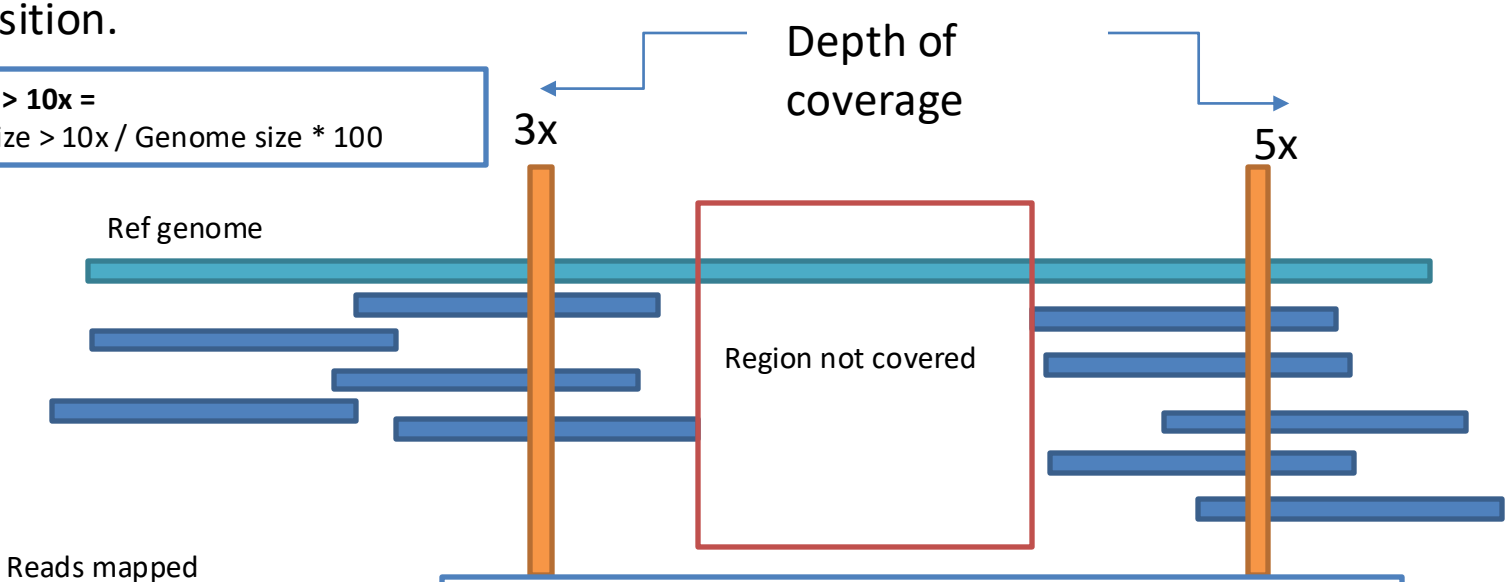
Mandatory parameter for microbial genomics!! It indicates us how many reads we have from our organism of interest. In human genomics this is almost always 99.99% unless something terrible happens. Not here!!!

# Mapping quality control

- **% genome > 10x:** percentage of genome covered with more than 10 reads.
- **Mean Depth of coverage:** mean of reads covering a genome position.

Picard
Samtools

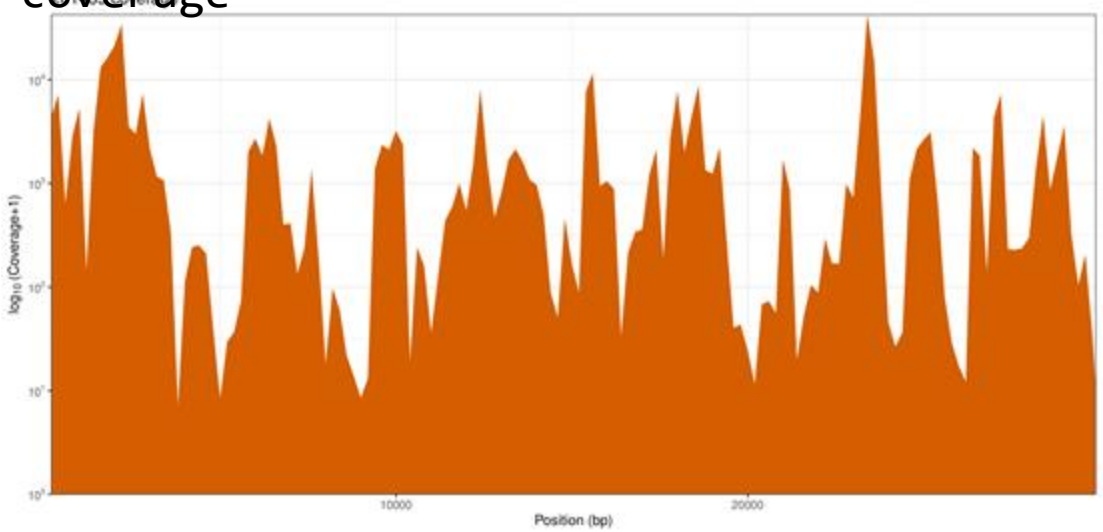
**% genome > 10x =**
Genome size > 10x / Genome size * 100



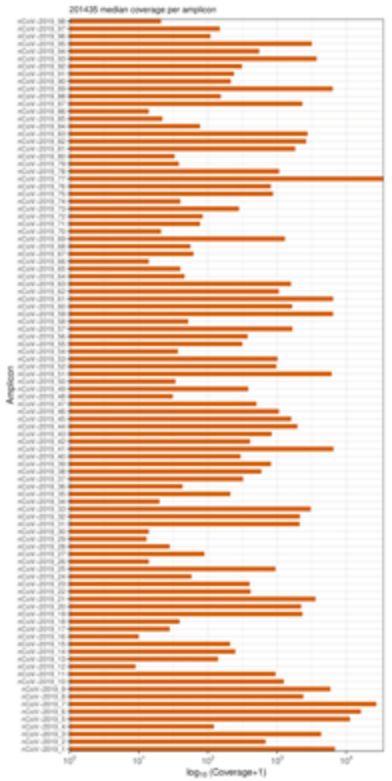

**Mean Depth of coverage =**
Depth of coverage in position 1, 2, 3 / Genome size

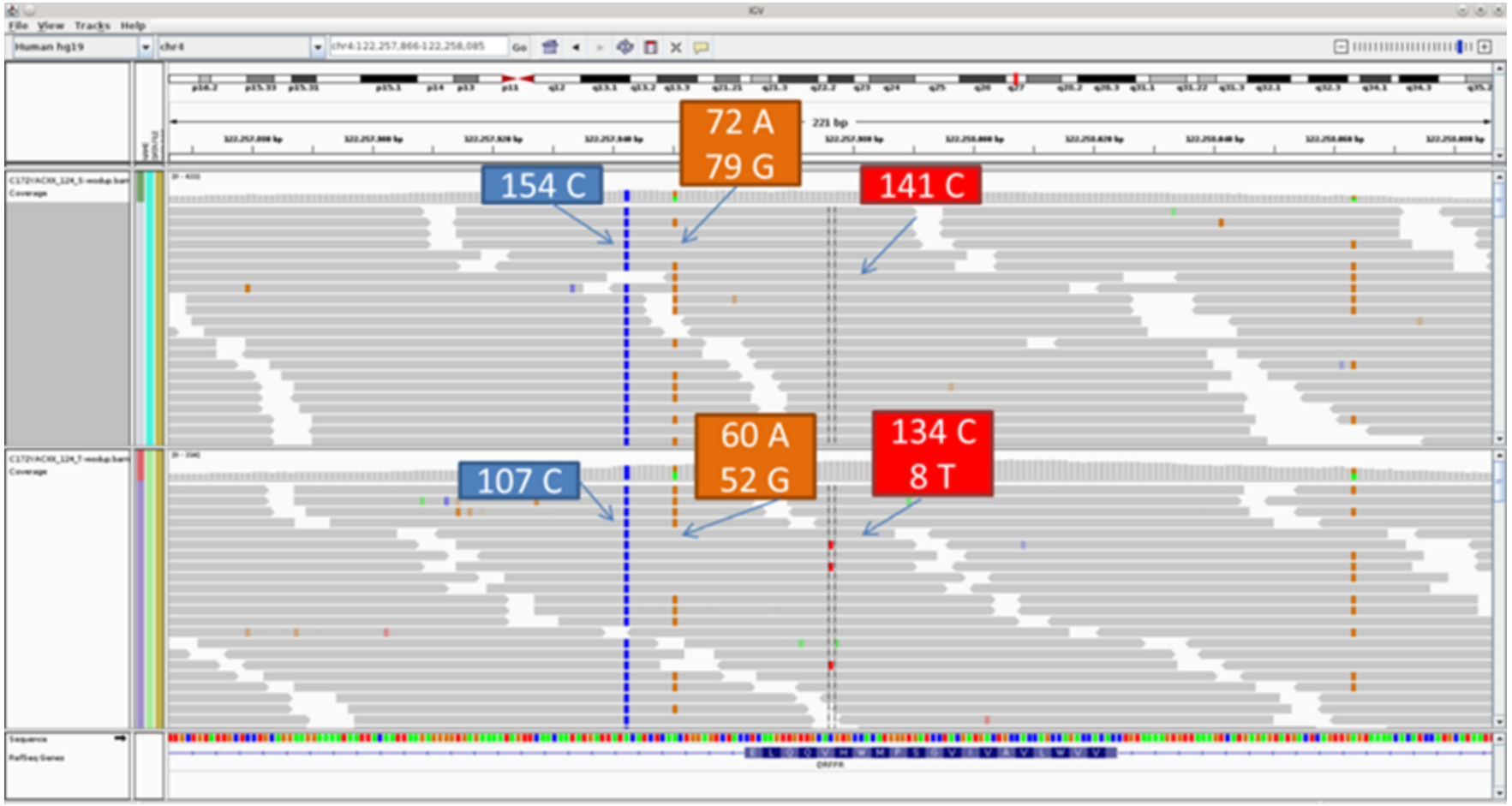
# Amplicon QC results

Genome coverage




Amplicon coverage




72 A
79 G
154 C
141 C
60 A
52 G
107 C
134 C
8 T

# Thanks for your attention!